

Examples of NPU-equipped processors used in edge devices as of 2025

Apple M4 with Neural Engine (16-core)

- ~38 TOPS (INT8) performance
- Optimised for on-device AI tasks like image recognition and natural language processing
- Low-latency enables real-time applications like live transcription and gesture detection
- Integrated directly into the SoC for power-efficiency
- Best for macOS/iOS ecosystems
- Used in devices like iPad Pro 2024, Mac mini m4, and MacBooks

Intel Core Ultra 200V (Lunar Lake) with NPU 4

- Up to 48 TOPS
- Integrated with Arc graphics
- Balances CPU/GPU/NPU for multitasking and battery life (~20+ hours video playback)
- Strong for Windows AI (Copilot+ certified); and used in Copilot+ PCs like Surface Laptop 7
- Also offers 200S (Arrow Lake) processors optimised for desktops; up to 36 TOPS

AMD Ryzen AI 300 (Strix Point) with XDNA 2 NPU

- Up to 50 TOPS
- Pairs with Radeon 890M iGPU
- Superior multi-threaded AI workloads
- Great value in premium laptops for creators and gamers using AI tools like Stable Diffusion
- Used in devices like the ASUS Zenbook S16 and the HP OmniBook

Qualcomm Snapdragon X Elite with Hexagon NPU

- 45 TOPS
- Arm-based architecture for top battery life (20+ hours)
- Optimised for Windows on ARM with Prism emulation
- Excels in mobile AI, like real-time translation and photo editing
- Used in devices like the Microsoft Surface Laptop 7, the Samsung Galaxy Book4 Edge

MediaTek Dimensity 9400 with APU 890 (7th-gen)

- Up to 80 TOPS (system-level AI)
- System-level AI processor for smartphones
- Handles GenAI features (e.g., video generation)
- Works well in mid-to-high-end Android devices
- Used in flagship Android phones like Vivo X200

(Generated with help from Yupp.ai)

models. These small processing units are capable of **massive parallelism** and can work on different parts of an AI model or dataset simultaneously, enabling rapid, concurrent execution of complex algorithms.

As AI inference does not require precision, NPUs generally **use lower-precision numbers** (8-bit or lower) instead of high-precision floating-point numbers used by CPUs and GPUs, to reduce computational complexity and thereby increase energy efficiency and speed, while reducing the memory footprint. To

prevent delays caused by moving data between external memory and the processor, most NPUs include **on-chip static random access memories (SRAMs)** like caches and scratchpad memory.

The arrangement of the processing elements and memory, and the optimisation of data and workflow, are critical to improving the performance of an NPU. The **performance of an NPU is measured in tera/trillion operations per second (TOPS)**. Modern NPUs boast of 35-50 TOPS. Some manufacturers mention the total system TOPS,

which includes the performance of the CPU and GPU, while others rate the NPU capability alone. When comparing options, make sure you are looking at the same measure.

Working in tandem

While large data centres may have standalone NPUs, in personal devices and other edge devices, the NPU is combined with other coprocessors, such as the CPU and GPU, into a single system-on-chip (SoC).

An AI PC's **firmware and customised operating system (OS)** will ensure that each processor does what it is cut out for, freeing up the others for their work. Since a lot of sensitive data is processed locally in AI devices, they usually have built-in security features in hardware, firmware, and BIOS, ensuring that threats are detected even before the OS boots. PCs handling AI workloads will also require a lot of RAM, with 16GB as the minimum and 32GB as the recommendation.

AI PCs usually come pre-loaded with a host of AI assistants and apps for tasks such as computer vision, video enhancement, noise cancellation, image editing, natural language processing (NLP), real-time translation, and gesture recognition. While the built-in suite is very helpful, those who invest in an AI PC would generally do much more with it—like running open AI models such as GPT-OSS or Llama 3 locally on the PC using tools like Ollama, LM Studio, or GPT4All, training and running their own AI agents, and so on.

Practically speaking, **serious AI users and developers will prudently use both AI PCs and the cloud to achieve their goals**—with the AI PC for real-time, on-device inference and/or small model training, and the cloud for larger, more complex model training. If you have, or foresee such intensive AI needs, an AI PC would be a worthy investment. But do look into the specifications to make sure it really is one! **EFY**